

1. Project Overview

1.1 Project Name

Hiring Assistant for HR & Recruiters

1.2 Project Idea (Business Perspective)

An innovative intelligent automation system specifically designed to help Human Resources (HR) professionals and companies overcome the traditional challenges of manually reviewing hundreds of resumes (CVs). The system handles application submissions, evaluates and filters candidates accurately based on specific job requirements and criteria, and sends detailed reports to the manager containing the top-qualified candidates to streamline the decision-making process for acceptance and scheduling interviews in the fastest and simplest way possible.

2. Problems Solved by the Project

Problem	How the Project Addresses It
Time and Effort Consumption	Eliminates the HR struggle of manually reading and filtering massive volumes of CVs for every open vacancy.
Human Error and Bias	Reduces human errors and unconscious biases when evaluating essential candidate skills and experiences.
Communication Delays	Prevents delays in responding to top talent, avoiding the loss of great candidates due to slow screening processes.
Poor Organization	Solves the scattering of candidate data across separate emails and files by providing a unified and clear reference for each job role.

3. Business Value

- Significantly accelerates the hiring process (Time-to-Hire).
- Guarantees the shortlisting of top talents who genuinely match job requirements with high accuracy driven by smart, numerical evaluation.
- Provides an organized and seamless workflow connecting application intake, evaluation, and recruitment task management in a single place.

4. Tech Stack

Component	Role in the System
n8n Form Trigger	Online application form that receives candidate details and CV files and starts the workflow.
Supabase	Cloud database storing job requirements and candidate applications.
Cohere Chat Model (command-r-plus-08-2024)	Language model powering the CV analysis.
AI Agent	Analyzes skills and experience, then generates the numerical score and candidate summary.
ClickUp	Used for task management to eliminate operational bottlenecks. Instead of stalling the queue (where a second candidate had to wait for HR to manually reply to the first), ClickUp enables parallel tracking and simultaneous processing for multiple applications without delays.
Gmail	Sends interactive emails to the HR manager and rejection emails to candidates.

5. Workflow Flow

- 1. Application Intake:** Candidate details and resume (CV) files are received through the online application form submission.
- 2. File Processing:** The contents of the candidate's PDF resume are processed and parsed to clearly extract the text.
- 3. AI Analysis:** The data is sent to the AI Agent powered by the Cohere model to analyze skills and experience, and to generate a numerical score and professional candidate summary.
- 4. Database Storage:** The evaluation results and candidate data are saved in the Supabase database table, linked with the job ID (`job_id`) to ensure relational integrity.
- 5. Smart Routing (Switch):** The evaluation result and score are assessed to route the application into two distinct paths based on company criteria:
 - **Acceptance Path (HR Notification):** For candidates who meet the requirements.
 - **Rejection Path (Rejection Email):** For candidates with low scores or those who do not meet the criteria. and their status is updated in the database.
- 6. Task Management & Communication (In case of Acceptance):**
 - An automated task is created inside ClickUp for the candidate to track recruitment steps.
 - An interactive email is sent to the HR manager containing candidate details along with direct buttons to accept or reject.

- 7. Automated Rejection (In case of Rejection):** An automated rejection email is sent to the candidate stating that their application was unsuccessful, and their status is updated in the database.

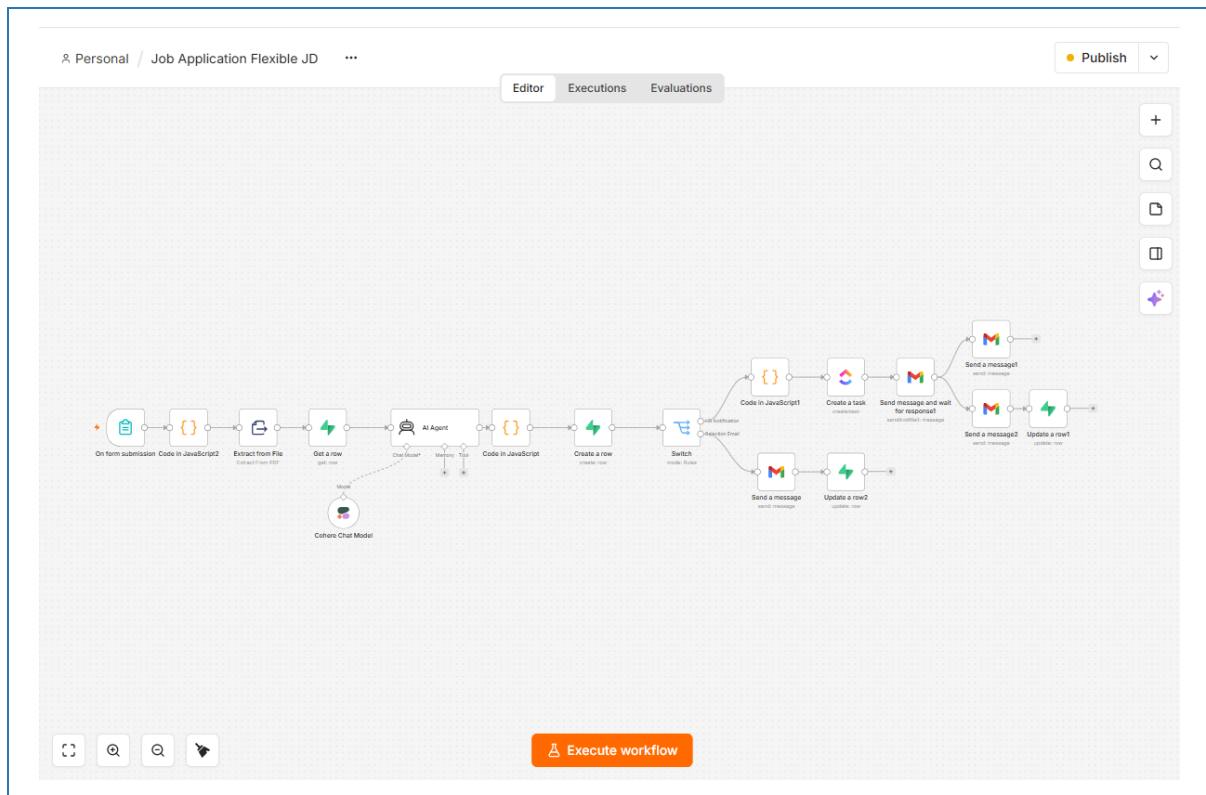


Figure 1: n8n workflow of the recruitment and screening system

6. Database Schema

The project's database architecture relies on an organized cloud-based system via Supabase to securely and systematically store job data, candidate information, and resume analyses. The structure consists of the following two main tables:

6.1 Job Requirements Table (job_requirements)

Table Concept: Represents the reference table for available vacancies in the company and the requirements for each job, against which resumes are evaluated.

Column	Type / Key	Description
id	Primary Key	The unique identifier for each job.
title	Text	The title of the open position (e.g. BI Developer, Data Analyst).
description	Text	The detailed job description and required skills for the position.

6.2 Candidate Applications Table (User_job_applications)

Table Concept: Dedicated to recording each candidate's personal data, comprehensive AI analysis results from their CV, system status, and the position they applied for.

Column	Type / Key	Description
id	Primary Key	The unique identifier for each application.
fname	Text	The full name of the candidate extracted from the application form or CV.
email	Text	The candidate's email address for communication.
score	Integer	The numerical score for the candidate extracted via the AI Agent based on skill matching with job requirements.
summary	Text	The professional summary of the candidate's evaluation and experience generated by AI.
status	Text	The current application status (e.g., Pending, Approved, Rejected).
links	Text	The candidate's professional links extracted from the CV (e.g., LinkedIn, GitHub, or portfolio).
language	Text	The languages spoken by the candidate (e.g., Arabic, English) extracted from their resume.
education	Text	The candidate's academic and educational background (e.g., university, major, or degree).
skills	Text	The technical and soft skills possessed by the candidate extracted from the CV.
courses	Text	The training courses and professional certifications obtained by the candidate.
experience	Text	Previous work experiences, employment history, and companies the candidate worked for.
job_id	Foreign Key	Links the application to the primary job ID in the job_requirements table to ensure relational integrity between jobs and their applicants.

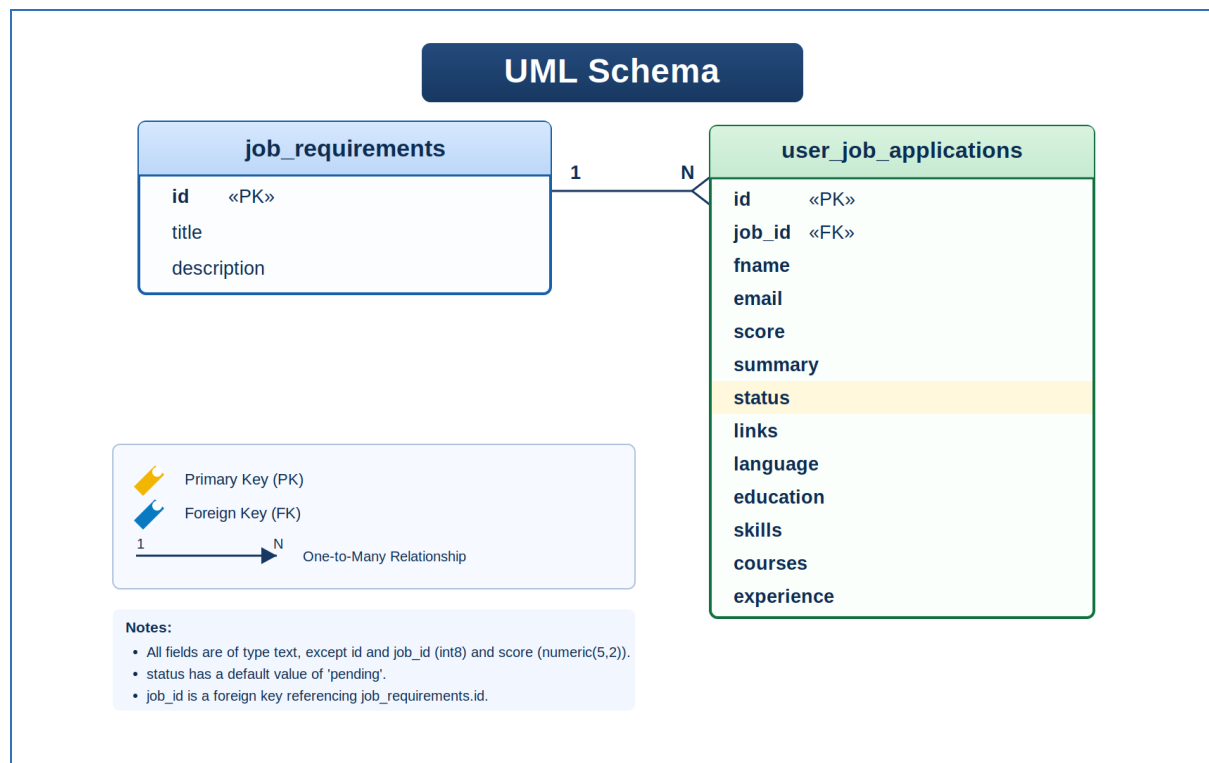


Figure 2: Database schema and table relationship in Supabase

7. Troubleshooting & Operational Updates

During the development and integration phase across various systems (n8n, Supabase, Cohere AI, Gmail, ClickUp), several technical challenges and errors were addressed, alongside operational improvements to enhance system efficiency.

7.1 Reasons for Using ClickUp (Business Perspective)

Problem : Before integrating ClickUp, the system relied on interactive email nodes that forced the HR manager to manually reply to the first candidate's email before the system could process the next applicant. This caused a severe operational bottleneck, leaving subsequent candidates waiting indefinitely without progress.

Solution (ClickUp's Role) : ClickUp was integrated to transform the follow-up process into parallel automated tasks. Notifications and tasks for multiple candidates are now dispatched simultaneously, ensuring that no application is blocked or waiting for the HR response of a preceding applicant, thereby drastically accelerating the entire recruitment workflow.

7.2 Operational Adjustments on Gmail Nodes

Adjustment: The email sending node for candidates (which previously went through Send and Wait for Response) was changed to a direct sending node (**Send only**).

Reason: Because automated acceptance emails do not require waiting for interaction or a response from the candidate (unlike the HR email, which requires interactive accept/reject buttons), direct sending speeds up the process and prevents the workflow from hanging while waiting for an unnecessary response.

7.3 Resolving Data Storage Issues (Null Values in Advanced Columns)

Problem: Language (language) and link (links) fields were returning NULL values inside the Supabase database, and standard text parsing failed to capture hyperlinks embedded inside PDF resumes generated via platforms like FlowCV.

Cause: The new columns were not mapped properly in the Supabase Insert Node, and raw text extraction alone cannot read hidden URI links inside PDF binaries.

Solution: A JavaScript Code snippet was added at the beginning of the workflow to read the PDF binary data as a buffer and scan it using a regular expression (URI Regex) to reliably extract any hidden hyperlinks into a detectedLinks variable.

The Supabase Insert Node configuration was updated to explicitly map all dynamic fields (languages, extracted links, skills, education, and courses), ensuring that all candidate data is successfully stored in the database without returning NULL values.

7.4 NULL Value in the job_id Field

Problem: While saving new application submissions in the User_job_applications table inside Supabase, the job_id foreign key field was returning a NULL value. This broke the relationship between the applicant and the vacant job, making it impossible to accurately match job requirements.

Cause: Failure to link the field to a compatible input variable in the Supabase Insert / Create Row Node within n8n.

Solution: The expression settings for the job_id field in the insert node were adjusted and connected to the id coming from the job requirements table query (job_requirements), automatically linking each application to the correct job ID.

7.5 Blank or Unformatted Messages in HR Notification Emails

Problem: Interactive emails sent to the Human Resources (HR) manager were arriving with empty fields (e.g., Email: / 100, AI Summary:) or displaying code snippets like {{ \$json.html }} in red text inside the message body.

Cause: Incorrect syntax when writing expressions or relying on static text without properly linking them to the output data coming from the AI Agent or Code Nodes.

Solution: The HTML body inside the Gmail node (Send message and wait for response) was restructured, and all dynamic fields (such as applicant name, email, score, and summary) were linked via direct outputs from the previous nodes using correct drag-and-drop styling to ensure data formatting and completion.

7.6 Adding the Extract from File Node

Problem / Reason: Since many resumes (CVs) are generated using platforms like **FlowCV**, their extracted contents do not natively output in a clean or raw text format that the AI Agent can easily parse directly.

Solution: An **Extract from File** node was specifically introduced right after the form submission and before the AI analysis. This ensures that the PDF file content is accurately parsed and converted into consistent raw text, which is then smoothly passed forward for analysis and data extraction.

8. Architectural Evolution & System Updates

During the course of developing and designing the system, several core improvements were implemented to enhance platform flexibility and user experience, most notably:

8.1 Shift from Static Job Description to Dynamic job_req Table

Before Modification	New Modification
The job description for vacancy requirements was static and hardcoded directly inside the AI prompt.	An independent table for job requirements (job_req) was created in the database, and a dynamic job data retrieval step (Supabase Insert/Get Row) was added inside the workflow, allowing vacancy details to be fetched based on the applicant's selection.

8.2 Updating AI Prompts (Dynamic AI Prompts & Drag & Drop)

Before Modification	New Modification
The prompt guided the AI based on fixed, standardized criteria for all applicants.	The prompt now relies on the Drag & Drop technique for data coming directly from the job_req table and the database node. This enables the AI to evaluate the CV specifically against the exact job description of the targeted position with supreme accuracy.

8.3 Updating the Application Form (Dynamic Form with Dropdown List)

Before Modification	New Modification
The application form was dedicated to a single position.	The form interface was upgraded to include an interactive dropdown list displaying currently available open positions. This allows applicants to easily select their desired job title, automatically passing the job_id to the remaining workflow steps.